

ESERCIZI

mercoledì 9 marzo 2022 17:55

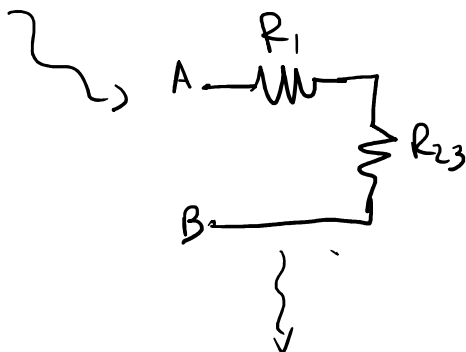


$$R_1 = 10 \, \Omega$$

$$R_2 = 1 \, \Omega$$

$$R_3 = 9 \, \Omega$$

TROVARE LA RESISTENZA
EQUIVALENTE AI TERMINALI
A e B. $R_{AB} = ?$



$$R_{23} = \left(\frac{1}{R_2} + \frac{1}{R_3} \right)^{-1} = \left(\frac{R_3 + R_2}{R_2 R_3} \right)^{-1} =$$

$$= \frac{R_2 R_3}{R_2 + R_3} = \frac{9}{10} \, \Omega = 0,9 \, \Omega$$

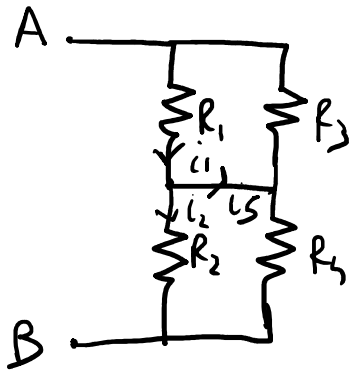
$$R_{23} \leq R_2, R_3$$

$$R_{AB} = R_1 + R_{23} = (10 + 0,9) \, \Omega = 10,9 \, \Omega$$

$$R_{AB} \geq R_1, R_{23}$$

Es. 2

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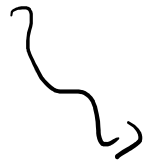
$$R_1 = 40 \text{ k}\Omega$$

$$R_2 = 10 \text{ k}\Omega$$

$$R_3 = 15 \text{ k}\Omega$$

$$R_4 = 40 \text{ k}\Omega$$

$$R_{AB} = ?$$



$$R_{13} = R_1 // R_3 = \frac{R_1 R_3}{R_1 + R_3} = 10,9 \text{ k}\Omega$$

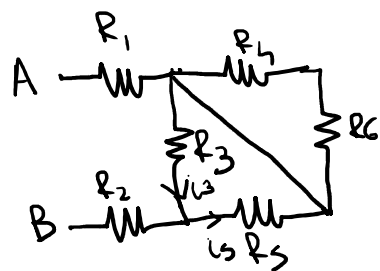
$$R_{24} = R_2 // R_4 = \frac{R_2 R_4}{R_2 + R_4} = 8 \text{ k}\Omega$$



$$R_{AB} = R_{13} + R_{24} = 18,9 \text{ k}\Omega$$

Es. 3

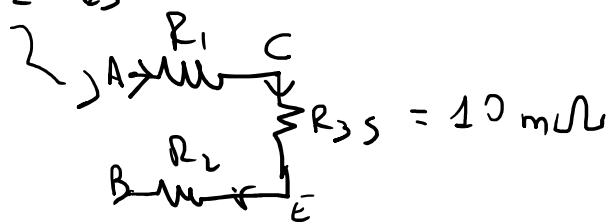
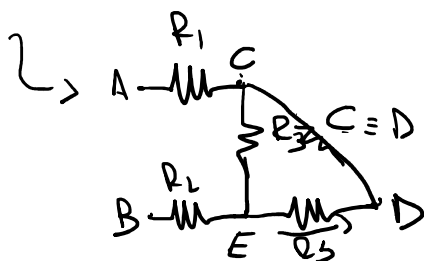
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$$R_1 = R_2 = 10 \text{ m}\Omega$$

$$R_3 = R_4 = R_5 = R_6 = 20 \text{ m}\Omega$$

$$R_{AB} = ?$$

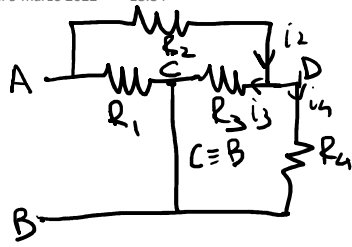


$$R_{AB} = 30 \text{ m}\Omega$$

$$R = R_4 // R_6 = \frac{R_4 R_6}{R_4 + R_6} = 0$$

E5, 4

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$$R_1 = 4 \text{ k}\Omega$$

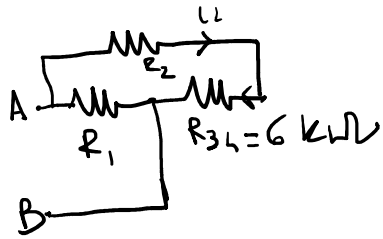
$$R_2 = 6 \text{ k}\Omega$$

$$R_3 = R_4 = 12 \text{ k}\Omega$$

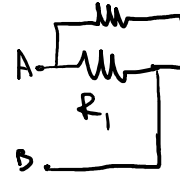
$$R_{AB} = ?$$

$$R_{234} = 12 \text{ k}\Omega$$

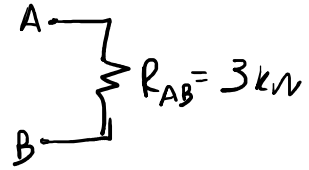
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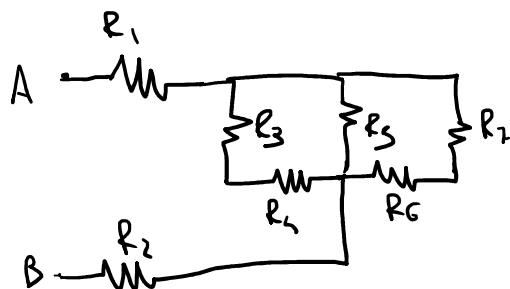


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Es. 5

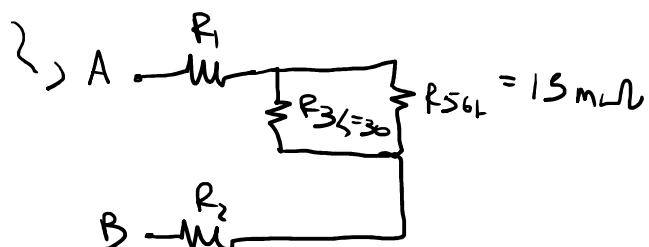
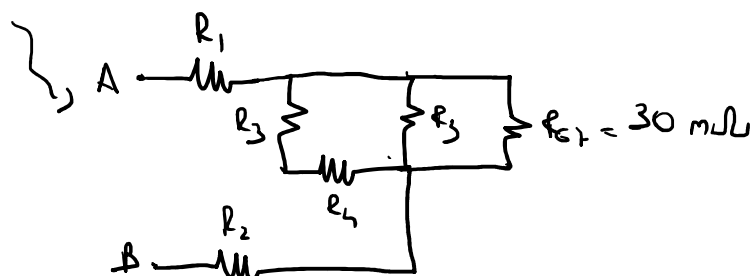
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$$R_1 = R_2 = R_3 = R_6 = 10 \text{ m}\Omega$$

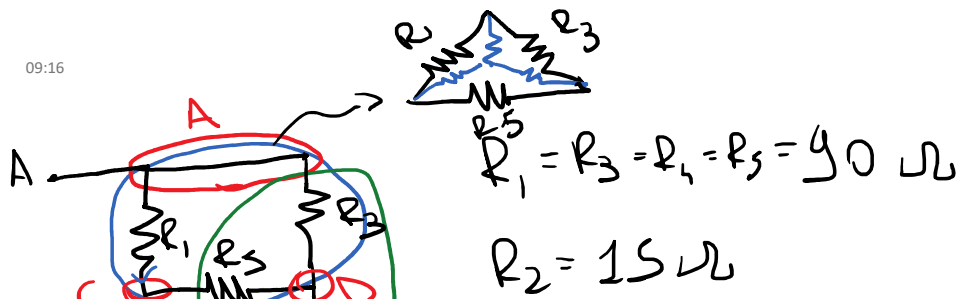
$$R_4 = R_7 = 20 \text{ m}\Omega$$

$$R_5 = 30 \text{ m}\Omega$$



Es.6

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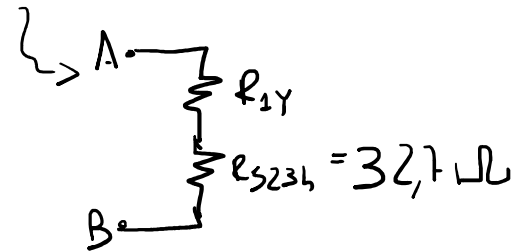
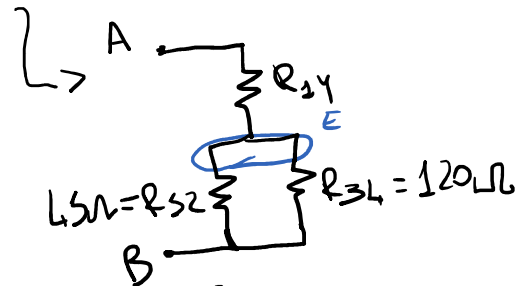
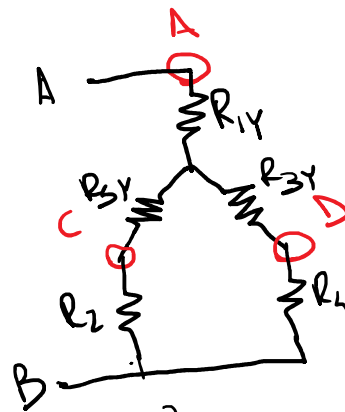


$$R_Y = \frac{1}{3} R_\Delta$$

$$R_{1Y} = \frac{1}{3} R_\Delta = 30 \Omega$$

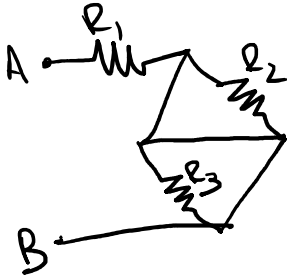
$$R_{5Y} = \frac{1}{3} R_\Delta = 30 \Omega$$

$$R_{3Y} = \frac{1}{3} R_\Delta = 30 \Omega$$



Es. 7

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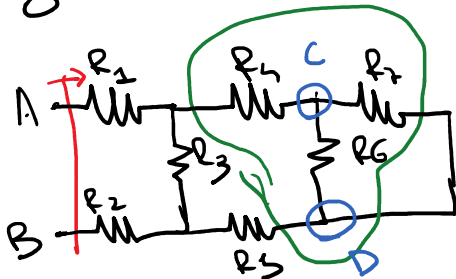
$$R_1 = 5 \Omega$$

$$R_2 = 12 \Omega$$

$$R_3 = 7 \Omega$$

$$R_{AB} = ?$$

Es. 8



$$R_1 = 2 \Omega$$

$$R_2 = 3 \Omega$$

$$R_3 = 18 \Omega$$

$$R_4 = 5 \Omega$$

$$R_5 = 8 \Omega$$

$$R_6 = R_7 = 10 \Omega$$

$$R_{AB} = ?$$

